Interval-Membership Preservation under Bounded Score Perturbations

Research Architecture Runtime

`operator/interval-membership`

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Abstract

`interval-membership` is a scalar region indicator preserved outside an ε-buffer under $|x' - x| \le \varepsilon$. Lean `LEAN_FULL`.

This theorem is: `Primitive`

1 Problem

The `interval-membership` operator maps a scalar score to a discrete region label/bit.

2 Stability notion

Additive $|x' - x| \le \varepsilon$; outputs agree when x is at least ε inside its region.

3 Definitions

Region boundaries are thresholds (sign at 0, abs-threshold at $T \ge 0$, interval $[L, U]$). Assumptions follow the Lean module.

4 Theorem

Theorem 1. *Under the module's buffer hypotheses, $|x' - x| \le \varepsilon$ preserves the operator output; sharpness pushes across a boundary when the buffer fails.*

5 Intuition

Same one-dimensional buffer logic as thresholding, applied to the operator's cut set.

6 Examples

Example 2. For sign: $x = 2$, $\varepsilon = 0.5$ stays positive. For abs-threshold $T = 3$, $x = 5$, $\varepsilon = 0.2$ stays pass.

7 Proof

Write $|x' - x| \leq \varepsilon$. On the pass side of each cut, x is at least ε beyond the cut, so x' cannot cross it; on the fail side, x' cannot reach the cut. Sharpness uses a $\pm\varepsilon$ push from a point within ε of the boundary (Theorem **??**), matching the Lean cases.

8 Formal statement

Lean module
`Research.Operators.IntervalMembership.Preservation`

Source file
`lean/Research/Operators/IntervalMembership/Preservation.lean`

Theorems

- `interval\_membership\_preservation`
- `interval\_membership\_sharpness`

Certificate
`lean/certificates/interval-membership/interval-membership-preservatio n`

Status
`LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`)

9 Proof dependencies

- Interval arithmetic on $\mathbb{R}$.
- Threshold-style buffers.

10 Consequences

Scalar building block; related: thresholding, multi-threshold, absolute-value-threshold.